

What Is Con

‘Convergence’ is a lot of things, and means different things to different people—as you’ll find out in the pages that follow

Few devices exemplify digital convergence better than the smartphone with its various functions—that of a PDA, cell phone, camera, and more. There was a market—now fast declining—for dedicated PDAs; and a huge market for cell phones. Someone tacked those two together, and the result was a huge success. And then, someone else threw in a camera as well, and that was the masterstroke. Smartphone sales are already eating into PDA and vanilla cell phone sales, and ABI Research (Allied Business Intelligence Research, New York, USA) predicts that by 2009, smartphones and connected PDAs will represent nearly a quarter of all handsets shipped.

In simple terms, this is what digital convergence is all about—the coming together of diverse technologies into simple devices that are more useful than the original individual devices themselves. But there’s more to convergence than just this. It’s an inexorable force that is driving markets worldwide; it’s a force that is pushing innovation to the limits; it’s a revolution.

Let’s get back to the clock—the simple time-displaying device. In days bygone, a wristwatch or pocket watch were solely how we told time. Then came the LCD and LED, and we had clocks all over the place—in those old Nintendo and Casio pocket video games, in home music systems, in TVs, in VCRs. This is a primitive, but very indicative, example of convergence. Think about it. It is, in fact, a very good example. What happened?

Essentially, manufacturers decided to throw in clocks into diverse devices, and people liked it—they, in fact, began to expect this of all electronics devices. You might ask why a video game should display time,

and that brings us back to the smartphone: who wants a camera in a phone? In fact, many do. Which brings us to the question: Was it the market that drove the innovation, or did the finished product create demand?

All we know as of now is that someone took a piece of duct tape and stuck a camera onto a phone (and perhaps we’ll soon see people throwing in phones into cameras!). The idea clicked (no pun intended) and as a result, a huge new market was created in a short span of time.

One driving factor here, could have been the fact that with a phone, which these days is equipped with Infrared and Bluetooth communications possibilities, people could share their photos at the press of a button unlike with other media.

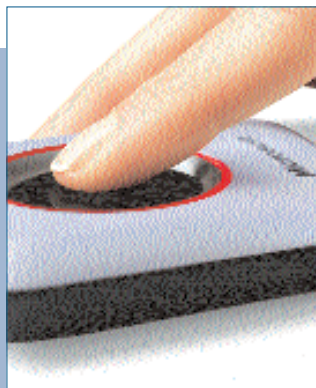
If markets drive convergent innovations, we have to ask, what are the ideal converged devices people want? And if innovations create demand, we have to ask, what of ideas that don’t work out?

As regards the former, think of the following as the ultimate example of digital convergence: your fridge automatically detects there’s not enough milk. Without your even having to press a button, it uses your home’s wireless network to get on the Internet, places an order at your local supermarket and the milk is delivered. Some call this the Holy Grail of digital convergence.

One needs to see what is at work here: it’s the same thing that drove almost all technological innovations—the idea of making human life easier. That is the essential driving force. But there’s more on the subject of what people want. With huge flat-screen TVs making their presence felt everywhere, courtesy dropping prices, they will become essential components of the networked home in the not-too-distant future. And people will want content to fill those TV screens with. That hunger for content will come simply as a result of the TVs being there. The fulfillment of one market will create another.

What about convergence ideas that didn’t take off? Sometimes, when one idea is shotgun-married to another, the result turns out to be something no one wants. A prime example is the CueCat—a mouse-like device that could scan bar codes, whose corresponding products you could look up on the Internet with one click. For reasons unknown, it flopped.

No one can pinpoint why exactly the CueCat flopped; think about an almost identical concept—a camera in a phone that can click a bar code. One can then log on to the Internet for information. Say you’re buying meat: the barcode contains information about how the animal was bred and reared, and you get this within a moment of scanning the barcode; this helps



Convergence is not just about gadgets. It’s also about connectivity (*left*) and the facilitation of everyday things, exemplified by the fingerprint scanner (*right*)